1. Zinc (65.37 g/mol) is determined by precipitating and weighing $Zn_2Fe(CN)_6$ (342.70 g/mol). What mass of zinc in grams is contained in a sample that gives 0.438 g precipitate?
   - a. 2.19
   - b. 0.167
   - c. 0.334
   - d. 2.30
2. What is the concentration in ppm of a 1.5×10−31.5×10−3 molar solution of NaCl (58.5 g/mol)?
   - a. 88 ppm
   - b. 8.8 ppm
   - c. 150 ppm
   - d. 1.5 ppm
3. Which one of the following precipitates is more soluble in its saturated solution?
   - a. AgCl (Ksp = 1.8×10^{-10})
   - b. AgI (Ksp = 8.5×10^{-17})
   - c. AgBr (Ksp = 5.0×10^{-13})
   - d. Ag_2CrO_4 (Ksp = 1.2×10^{-12})
4. Given that the solubility product for $La(IO_3)_3$ is 1.0×10−111.0×10−11, what is the concentration of La^{3+} in a saturated solution of lanthanum iodate?
   - a. 1.35×10−31.35×10−3 M
   - b. 1.0×10−31.0×10−3 M
   - c. 7.18×10−57.18×10−5 M
   - d. 7.8×10−47.8×10−4 M
5. How many grams of 2.50% by mass aqueous HF (20.0 g/mol) are required to provide 20% excess to react with 50.0 mL of 0.0325 M Th^{4+} by the reaction:
   $Th^{4+} + 4\,F^- \rightarrow ThF_4\,(s)$
   - a. 3.12
   - b. 25.0
   - c. 6.24
   - d. 12.5

6. At 700 K, the reaction 2SO2(g)+O2(g)⇌2SO3(g)$2SO2(g)+O2(g)\rightleftharpoons 2SO3(g)$ has the equilibrium constant Kc=4.3×106$Kc=4.3\times 106$, and the following concentrations are present: [SO2]=0.10 M; [SO3]=10. M; [O2]=0.10 M$[SO2]=0.10M;[SO3]=10.M;[O2]=0.10M$.
   Is the mixture at equilibrium? If not, in which direction (as the equation is written), left to right or right to left, will the reaction proceed to reach equilibrium?
   - a. There is not enough information to predict the direction
   - b. No, right to left
   - c. No, left to right
   - d. Yes, the mixture is at equilibrium
7. A student performs an experiment to determine the density of a sugar solution. She obtains the following results: 4.71 g/mL, 4.73 g/mL, 4.67 g/mL, 4.69 g/mL. If the actual value for the density of the sugar solution is 4.40 g/mL, which statement below best describes her results?
   - a. Her results are both precise and accurate
   - b. Her results are precise, but not accurate
   - c. Her results are accurate, but not precise
   - d. It isn't possible to determine with the information given
8. How many milliliters of 0.260 M Na_2S are needed to react with 40.00 mL of 0.315 M $AgNO_3$?
   $Na_2S(aq) + 2\ AgNO_3(aq) \rightarrow 2\ NaNO_3(aq) + Ag_2S(s)$
   - a. 12.1 mL
   - b. 66.0 mL
   - c. 24.2 mL
   - d. 48.5 mL
9. Determination of the sodium level in separate portions of a blood sample by ion-selective electrode measurement gave the following results: 139.2, 139.8, 140.1, and 139.4 meq/L. What is the range within which the true value falls, assuming no determinate error at the 99% confidence level (t at 99% confidence level = 5.84)?
   - a. 139.6±0.6$139.6\pm 0.6$
   - b. 139.6±0.5$139.6\pm 0.5$
   - c. 139.6±1.5$139.6\pm 1.5$
   - d. 139.6±1.2$139.6\pm 1.2$
10. What concentration of Ag^+ is required to precipitate ONLY AgBr in a solution that contains both Br^- and Cl^- at a concentration of 0.040 M?
    KspKsp of AgCl = 1.6×10−10$1.6\times 10-10$ and KspKsp of AgBr = 7.7×10−13$7.7\times 10-13$

- a. 1.0×10−11 M<[Ag+]<2.0×10−9 M1.0×10−11M<[Ag+]<2.0×10−9M
- b. 1.9×10−11 M<[Ag+]<4.0×10−9 M1.9×10−11M<[Ag+]<4.0×10−9M
- c. 1.0×10−11 M<[Ag+]<2.0×10−9 M1.0×10−11M<[Ag+]<2.0×10−9M
- d. 3.9×10−11 M<[Ag+]<8.0×10−9 M3.9×10−11M<[Ag+]<8.0×10−9M

11. Which of the following statements is most correct?
    - a. Disturbance of an equilibrium system results in a shift of the equilibrium position
    - b. Adding more reactants for a reaction at equilibrium decreases the value of K
    - c. Always salts with the same value of Ksp*Ksp* have similar solubility
    - d. At equilibrium, no more reactants are transformed into products
12. How many significant figures are there in the answer for the following calculation? (143.7−121)×2.060.6000.600(143.7−121)×2.06
    - a. Three
    - b. Four
    - c. Two
    - d. One
13. The following replicates are obtained for the analysis of calcium in drinking water: 23.4, 23.2, 22.8, 26.0, 24.0, and 23.0 ppm. After applying the Grubbs Test, the mean is --- and the standard deviation is ---. Use Gtable=1.822*Gtable*=1.822.
    - a. 23.7 and 1.2
    - b. 23.7 and 0.5
    - c. 23.3 and 1.2
    - d. 23.3 and 0.5
14. What is the molar solubility of Hg2Cl2*Hg*2*Cl*2 (Ksp=1.2×10−18*Ksp*=1.2×10−18) in 0.060 M KCl?
    - a. 2.6×10−15 M2.6×10−15M
    - b. 1.3×10−15 M1.3×10−15M
    - c. 6.6×10−16 M6.6×10−16M
    - d. 3.3×10−16 M3.3×10−16M

Q1	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**	**Q11**	**Q12**	**Q13**	**Q14**
B	A	D	A	C	C	B	C	A	B	A	A	A	D

15. What is the molar solubility of AgBr in (a) pure water?
    $AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)$
    Ksp=7.7×10−13*Ksp*=7.7×10−13

    - a. 8.0×10−78.0×10−7
    - b. 9.2×10−89.2×10−8
    - c. 5.7×10−85.7×10−8
    - d. 8.8×10−78.8×10−7 M

16. What is the equilibrium constant for reaction (2), if the equilibrium constant for reaction (1) is 1.8×10−51.8×10−5?
    Reaction (1): NH3(aq)+H2O→NH4++OH−NH3(aq)+H2O→NH4++OH−
    Reaction (2): NH4++OH−→NH3(aq)+H2ONH4++OH−→NH3(aq)+H2O

    - a. 5.6×10−15.6×10−1
    - b. 5.6×1045.6×104
    - c. 5.6×10−45.6×10−4
    - d. 5.6×10−15.6×10−1

17. The ionic strength of a solution containing 0.1 M $FeCl_3$, 0.2 M KCl, and 1 M NH_3 is:

    - a. 0.9
    - b. 0.4
    - c. 0.8
    - d. 0.6

18. What volume of 25.0 wt% $AgNO_3$ (170 g/mol, density = 1.4 g/mL) is required to precipitate all chloride Cl^- (35.5 g/mol) in a 100 mL 0.25 M solution and give a 5% excess?
    $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + Na^+ + NO_3^-$

    - a. 14.5 mL
    - b. 12.7 mL
    - c. 12.5 mL
    - d. 12.1 mL

19. One part per million (ppm) is equivalent to one:

    - a. mg/g
    - b. ng/g
    - c. μg/g

- d. µg/kg

20. The following replicates are obtained for the analysis of calcium in drinking water: 23.4, 23.2, 22.8, 26.0, 24.0, and 23.0 ppm. After applying the Grubbs Test, the mean is --- and the standard deviation is ---. Use Gtable=1.822$Gtable=1.822$.
    - a. 23.3 and 0.5
    - b. 23.3 and 1.2
    - c. 23.7 and 1.2
    - d. 23.7 and 0.5
21. Choose the correct answer with the correct significant figures: (12)+(3.2)−(1.24)=?$(12)+(3.2)-(1.24)=?$
    - a. 14.0
    - b. 13.96
    - c. 14.16
    - d. 14
22. The measured pH of 0.0100 M HCl + 0.0900 M KCl at 25°C is 2.102. From this information, calculate the activity coefficient of H^+ in this solution.
    - a. 0.791
    - b. 1.00
    - c. 1.05
    - d. 0.890
23. You have 505 mL of 0.150 M HCl solution, and you want to dilute it to exactly 0.100 M. How much water would you add?
    - a. 1009 mL
    - b. 126 mL
    - c. 757 mL
    - d. 252 mL
24. The wt% and the ppm concentration of a solution made by dissolving 10.0 g salt in 240 g deionized water (density of water = 1.0 g/mL) are:
    - a. 4%, 40000 ppm
    - b. 2%, 2000 ppm
    - c. 1.66%, 16600 ppm

- d. 4%, 4000 ppm

25. Convert 5.5 Tm (terameter) to km:
    - a. 5.5×10155.5×1015
    - b. 5.5×1095.5×109
    - c. 5.5×10125.5×1012
    - d. 5.5×1065.5×106

Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25
D	B	C	B	C	B	A	A	D	A	B

26. The pH (-log[H^+]) of a solution is 4.164. The [H^+] is:
    - a. 6.85×10−56.85×10−5 M
    - b. 7.0×10−57.0×10−5 M
    - c. 6.8×10−56.8×10−5 M
    - d. 5.850×10−55.850×10−5 M
27. The density of a 5.24 M $NaHCO_3$ (84.0 g/mol) solution is 1.19 g/mL. Its molality is:
    - a. 6.99
    - b. 7.27
    - c. 9.97
    - d. 6.71
28. Suppose that $MgBr_2$ dissolves in pure water to give Mg^{2+}, Br^-, and $MgBr^+$. The mass balance for Br^- in 0.15 M aqueous solution of $MgBr_2$ is:
    - a. 0.45 M = 2[Br^-] + [$MgBr^+$]
    - b. 0.30 M = [Br^-] + [$MgBr^+$]
    - c. 0.45 M = [Br^-] + [$MgBr^+$]
    - d. 0.30 M = 2[Br^-] + [$MgBr^+$]
29. How many grams of $CuSO_4 \cdot 5H_2O$ (249.68 g/mol) should be dissolved in 250 mL to make 4.00 mM Cu^{2+}?

- a. 0.5
- b. 0.999
- c. 0.249
- d. 0.499

30. [log3.4×10−4]+4.45=?[log3.4×10−4]+4.45=?
    - a. 1.00
    - b. 0.98
    - c. 0.95
    - d. 0.99

31. How many grams of 0.491 wt% aqueous HF (20.0 g/mol) are required to provide a 50% excess to react with 25.0 mL of 0.0236 M Th^{4+} by the reaction:
    $Th^{4+} + 4F^{-} \rightarrow ThF_4(s)$
    - a. 1.8
    - b. 9.61
    - c. 14.4
    - d. 7.2

32. The $[H^{+}]$ of a solution is 2.70×10−62.70×10−6. What is the pH of the solution?
    - a. 5.5686
    - b. 5.57
    - c. 5.569
    - d. 44.48

33. What is the pH at which $[CO_3^{2-}] = 2[HCO_3^{-}]$?
    Given:
    H2CO3→H++HCO3−H2CO3→H++HCO3−
    Ka1=3.0×10−4*Ka*1=3.0×10−4
    HCO3−→H++CO32−HCO3−→H++CO32−
    Ka2=2.2×10−7*Ka*2=2.2×10−7
    H2O→H++OH−H2O→H++OH−
    - a. 5.9
    - b. 6.96
    - c. 4.4
    - d. 7.8

34. What is the molar solubility of AgBr in 0.0010 M NaBr?
    $AgBr(s) \rightleftharpoons Ag^{+}(aq) + Br^{-}(aq)$
    Ksp=7.7×10−13*Ksp*=7.7×10−13

    - a. 8.8×10−78.8×10−7
    - b. 3.5×10−73.5×10−7
    - c. 7.7×10−107.7×10−10 M
    - d. 3.57.8×10−73.57.8×10−7

35. The following data are for the determination of lead in gasoline: 5.2, 6.8, 6.5, 6.3, and 5.9. What is the true value at 90% confidence level? (t-table at 90% CL = 2.132).

    - a. 6.2±0.596.2±0.59
    - b. 6.1±0.66.1±0.6
    - c. 6.14±0.596.14±0.59
    - d. 6.1±0.56.1±0.5

36. Choose the correct statement:

    - a. Molarity is temperature-dependent
    - b. Molarity is always higher than molality
    - c. Molality is temperature-dependent
    - d. Molality is always higher than molarity

Q26	**Q27**	**Q28**	**Q29**	**Q30**	**Q31**	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**
A	B	A	D	B	D	B	B	C	C	A